

The GENIE–INCL quasi-elastic vertex in formulas: scattering frame and INCL energy balance

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Companion to `README.md` in `incl-potential-test/`; the code is `incl_nucleus.py` (ground state, potential, local energy) and `ep_incl_scatter.py` (scattering and balance). Every formula below is what the scripts evaluate, and every “verified” number is the maximum deviation over the 200k events of the named run (`ep_incl_C12_1f{on,never}[_resample].npz`, 2026-09-11). Sources: the INCL++ tree vendored in the `genie_inclxx` install and the fork branch `feature/incl-vertex-local-energy` at `6bd7803d6`; the convention is that of `docs/incl-vertex-local-energy-option-plan.md` (“Convention revised”).

Units MeV, MeV/c, fm. A primed symbol is an outgoing particle; a subscript or superscript 0 marks the scattering frame (before the balance); $\vec{x}$ is a 3-vector and $|\vec{x}|$ its magnitude.

1 Notation

symbol	meaning	C12 value
m	INCL nucleon mass (protons and neutrons)	938.2796
m_e	electron mass (<code>shared/pdg.json</code>)	0.511
p_F	Fermi momentum, $1.37 \hbar c (2Z/A)^{1/3}$	270.34
T_F	$\sqrt{p_F^2 + m^2} - m$	38.17
S	INCL separation energy	6.83
V_0	well depth $T_F + S$	45.00
$R_{\max}$	$5.5 + 0.3 (A - 6)/12$	5.65
$k = (E_e, \vec{k})$	beam electron, along $+z$	$E_e = 2445$
$P = (E_N, \vec{p}_N)$	the nucleon handed to the scattering	section 3
k', p'	outgoing electron and proton, final (after the balance)	
k'_0, p'_0	the same before the balance (scattering frame)	

2 INCL ground state

Density (modified harmonic oscillator, $6 < A \leq 19$):

$$\rho(r) \propto (1 + \alpha x^2) e^{-x^2}, \quad x = r/a, \quad 0 \leq r \leq R_{\max},$$

with the HFB parameters $a = 1.72905$ fm, $\alpha = 0.849882$ (C12 protons) for the default fuzzy correlation, and the medium-table values $a = 1.72$ fm, $\alpha = 0.84$ for a strict correlation.

r–p correlation. INCL does not draw r from ρ . It builds the density of *reflection radii*

$$g(R) = -R^3 \frac{d\rho}{dR} = 2R^2 x^2 (1 - \alpha + \alpha x^2) e^{-x^2} \quad (\text{ModifiedHarmonicOscillatorRP}),$$

$$F(R) = \int_0^R g(R') dR' / \int_0^{R_{\max}} g(R') dR',$$

stores the inverse table $R(u)$ with $u = F^{1/3}$ at 60 equidistant R nodes (linear interpolation, clamped at both ends), and its inverse $u(r)$. Then

$$\begin{aligned} \text{max_r_from_p}(x) &= R(x) && \text{reflection radius of a nucleon with } |\vec{p}| = x p_F, \\ \text{min_p_from_r}(r) &= u(r) && p_{\min}(r)/p_F: \text{ the smallest momentum that reaches } r, \\ p_{\min}(r) &= p_F F(r)^{1/3}. \end{aligned}$$

Strict sampling (coefficient ≥ 0.99999):

$$\begin{aligned} \vec{p}_{\text{ball}} &\text{ uniform in the ball } |\vec{p}| \leq p_F \quad ((p/p_F)^3 \text{ uniform}), \\ R &= R(|\vec{p}_{\text{ball}}|/p_F), \\ \vec{r} &\text{ uniform in the ball } |\vec{r}| \leq R, \\ p_{\text{refl}} &= |\vec{p}_{\text{ball}}| \quad (\text{reflection momentum}). \end{aligned}$$

Uniformly filled nested spheres with $dN/dR = g(R)$ generate

$$\rho_{\text{gen}}(r) = \int_r^{R_{\max}} g(R) \frac{3}{4\pi R^3} dR \propto \rho(r) - \rho(R_{\max}),$$

i.e. the MHO tail cut sharply at $R_{\max}$ (6.6 % low at 4.8 fm, 80 % low in the last 0.1 fm). Verified: sampled r against $r^2[\rho(r) - \rho(R_{\max})]$, $\chi^2/\text{ndf} = 0.6\text{--}1.3$ (400k nucleons, 40 bins, strict/fuzzy $\times 60/3000$ nodes).

Fuzzy sampling (coefficient $c = 0.5$ for protons, 0.73 for neutrons):

$$\begin{aligned} g_1, g_2 &\sim N(0, 1), & x_g &= g_1, & y_g &= c x_g + g_2 \sqrt{1 - c^2}, & u_1 &= \Phi(x_g), & u_2 &= \Phi(y_g), \\ x &= u_1^{1/3}, & y &= u_2^{1/3}, \\ |\vec{p}_{\text{ball}}| &= y p_F \text{ (isotropic)}, & R &= R(x), & \vec{r} &\text{ uniform in } |\vec{r}| \leq R, & p_{\text{refl}} &= x p_F. \end{aligned}$$

The marginals of r and p are those of the strict draw; only the joint correlation is loosened ($\text{corr}(p, r) = +0.34$ instead of $+0.68$), and 22 % of the nucleons lie below $p_{\min}(r)$.

The fork's redraw at fixed r (`ResamplingHitNucleon`, flag `--resample`; what GENIE's event loop feeds the vertex):

$$u = \text{min_p_from_r}(r), \quad |\vec{p}_{\text{ball}}|^3 \text{ uniform on } [u^3 p_F^3, p_F^3], \quad \text{isotropic}, \quad p_{\text{refl}} = |\vec{p}_{\text{ball}}|,$$

so $p_{\min}(r) \leq |\vec{p}_{\text{ball}}| \leq p_F$ ($\text{corr}(p, r) = +0.47$, $\langle p \rangle = 225.5$ MeV/c). In every case $E_{\text{ball}} = \sqrt{p_{\text{ball}}^2 + m^2}$ and $T_{\text{ball}} = E_{\text{ball}} - m \leq T_F$.

3 Potential energy and local energy

Potential (`NuclearPotentialEnergyIsospin`, $\alpha_V = 0.223$):

$$V(T) = \begin{cases} V_0, & T < T_F, \\ \max\left(0, V_0 - \frac{\alpha_V}{1 - \alpha_V} (T - T_F)\right) = \max\left(0, V_0 - 0.287 (T - T_F)\right), & T \geq T_F. \end{cases}$$

$V = 0$ above $T \approx 195$ MeV; every ball nucleon has $V(T_{\text{ball}}) = V_0$.

Local energy (`KinematicsUtils::getLocalEnergy`) for a nucleon at radius r with momentum p and reflection momentum p_{refl} :

$$p_{f0} = \begin{cases} p_F, & T \leq T_F, \\ \sqrt{t_{f0}(t_{f0} + 2m)}, & t_{f0} = V(T) - S, \quad T > T_F \quad (v_{\text{loc}} = 0 \text{ if } t_{f0} < 0 \text{ or } r > R_{\text{max}}), \end{cases}$$

$$p_l = p_{f0} \min_p_from_r\left(r \frac{R(p/p_{f0})}{R(p_{\text{refl}}/p_{f0})}\right),$$

$$v_{\text{loc}}(r, p) = \sqrt{p_l^2 + m^2} - m.$$

For a ball nucleon with $p_{\text{refl}} = |\vec{p}|$ (strict draw, or after the redraw) the ratio is 1 and $v_{\text{loc}} = T_F(r) \equiv \sqrt{p_{\text{min}}(r)^2 + m^2} - m$: 0 at the centre, T_F at R_{max} . For a fuzzy nucleon the ratio $R(y)/R(x)$ guarantees $v_{\text{loc}} \leq T_{\text{ball}}$ (0 clipped nucleons of 200k).

Local frame (fork `getHitNucleonP4`; `transformToLocalEnergyFrame` without mutation):

$$\begin{aligned} \text{on:} \quad E_{\text{loc}} &= \max(E_{\text{ball}} - v_{\text{loc}}, m), & \vec{p}_{\text{red}} &= \sqrt{E_{\text{loc}}^2 - m^2} \hat{p}_{\text{ball}} \quad (\text{on shell}), \\ \text{never:} \quad E_{\text{loc}} &= E_{\text{ball}}, & \vec{p}_{\text{red}} &= \vec{p}_{\text{ball}} \quad (v_{\text{loc}} \equiv 0), \end{aligned}$$

and $P = (E_{\text{loc}}, \vec{p}_{\text{red}})$ is the nucleon handed to the scattering.

4 Scattering frame (before the balance)

Two-body phase space on P : isotropic in the CM of $k + P$, then boosted to the lab, so the 4-momentum of $e + P$ is conserved exactly:

$$s = (k + P)^2, \quad p^* = \frac{\sqrt{[s - (m_e + m)^2][s - (m_e - m)^2]}}{2\sqrt{s}}, \quad k + P = k'_0 + p'_0.$$

Missing quantities of this state (the green curves of the figures):

$$\begin{aligned} \omega_0 &= E_e - E'_0 = E_{p'0} - E_{\text{loc}}, & \vec{q}_0 &= \vec{k} - \vec{k}'_0 = \vec{p}'_0 - \vec{p}_{\text{red}}, \\ E_{\text{miss}}^{(0)} &= \omega_0 - T_{p'0} = m - E_{\text{loc}} = -T_{\text{red}} \quad (\text{never: } -T_{\text{ball}}), & \vec{p}_{\text{miss}}^{(0)} &= \vec{p}'_0 - \vec{q}_0 = \vec{p}_{\text{red}} \quad (\text{never: } \vec{p}_{\text{ball}}). \end{aligned}$$

Verified: $|E_{\text{miss}}^{(0)} + T_{\text{red}}| \leq 2.4 \times 10^{-12}$ MeV and $||\vec{p}_{\text{miss}}^{(0)}| - |\vec{p}_{\text{red}}|| \leq 1.8 \times 10^{-12}$ MeV/c in all four runs. No binding appears here: the scattering sees an on-shell nucleon whose kinetic energy is T_{red} (about 0 at the surface with local energy on).

5 INCL energy balance

INCL's `InteractionAvatar` rule, as the fork implements it in `G4INCLGENIEAvatar`: `preInteraction` sets the conserved total, the `ViolationLeptonEMomentumFunctor` rescales the products, and INCL's `RootFinder` solves for the scale factor.

1. Conserved total, $E - V$ of the *global* nucleon, with no local-energy term: $E_{\text{tot}} = E_e + E_{\text{ball}} - V_0$.
2. Boost vector, the CM of $e + \text{local-frame nucleon}$: $\vec{\beta} = (\vec{k} + \vec{p}_{\text{red}})/(E_e + E_{\text{loc}})$.
3. Boost k'_0, p'_0 by $-\vec{\beta}$ to $\vec{k}_0^*, \vec{p}_0^*$ (with $\vec{k}_0^* + \vec{p}_0^* = 0$); scale both by one common factor, $\vec{k}^* = \alpha \vec{k}_0^*, \vec{p}^* = \alpha \vec{p}_0^*$, energies on shell $E^* = \sqrt{|\vec{p}|^2 + \text{mass}^2}$; boost back by $+\vec{\beta}$ to k', p' .
4. The outgoing proton takes its energy-dependent potential $V_{p'} = V(T_{p'})$. With local energy on, and only for slow protons ($V_{p'} \geq S$, otherwise $v_{\text{loc}} = 0$), the fixed point $E_{p'} \leftarrow E_{p'} + v_{\text{loc}}(r, |\vec{p}'|)$ with $|\vec{p}'|$ re-adjusted on shell, iterated to $|\Delta v_{\text{loc}}| < 10^{-4}$ MeV (`scaleParticleMomenta`).
5. Solve for α :

$$E' + E_{p'} - V_{p'} = E_e + E_{\text{ball}} - V_0$$

(bisection to $|\Delta E| < 2 \times 10^{-12}$ MeV here; INCL's `RootFinder` stops at 10^{-4} MeV).

α is 0.980–1.014 (mean 0.988 on, 0.983 never); no bracketing failures in 800k events.

Record (what GENIE writes): initial nucleon ($\vec{p}_{\text{ball}}, E_{\text{ball}} - V_0$), lepton k' , pre-FSI proton p' . Then

$$\omega = E_e - E', \quad \vec{q} = \vec{k} - \vec{k}', \quad E_{\text{miss}} = \omega - T_{p'}, \quad \vec{p}_{\text{miss}} = \vec{p}' - \vec{q} = \vec{k}' + \vec{p}' - \vec{k}.$$

6 Surface exit: from the well to a free proton

When the proton reaches the surface, `SurfaceAvatar` decides between transmission and reflection and `TransmissionChannel::particleLeaves` turns it into a free particle (default `refraction = false`). The kinetic energy outside is

$$\begin{aligned} T_{\text{out}} &= T_{p'} - V(T_{p'}) + \Delta Q, \\ \Delta Q &= [M_t(A, Z) - M_t(A - 1, Z - 1) - m_t] - [M_i(A, Z) - M_i(A - 1, Z - 1) - m] \\ &= -S_p^{\text{real}} + S = -15.957 + 6.83 = -9.127 \text{ MeV}, \end{aligned}$$

where M_t are the real masses, $A \text{ amu} + \Delta - Z m_e$ from `walletlifetime.dat` ($\Delta(^{12}\text{C}) = 0, \Delta(^{11}\text{B}) = 8.6679 \text{ MeV}$), $m_t = 938.27203 \text{ MeV}$ the table proton mass, and $M_i = Z(m - S) + (A - Z)(m - S)$ the INCL masses (`Particle::getEmissionQValueCorrection`): the real separation energy replaces

INCL's constant S at emission. Then

exits: $T_{\text{out}} > 0$ (otherwise reflection; $T_{p'} \lesssim 50$ MeV here),

$$P_T = \frac{4 p_{\text{in}} p_{\text{out}}}{(p_{\text{in}} + p_{\text{out}})^2} \times \begin{cases} 1, & T_{\text{out}} \geq B, \\ e^{-2L}, & T_{\text{out}} < B, \end{cases} \quad p_{\text{out}} = \sqrt{T_{\text{out}}(T_{\text{out}} + 2m)},$$

$$B = \frac{e^2(Z-1)}{R_p + 0.88 \text{ fm}} = 2.15 \text{ MeV} \quad (\text{Coulomb barrier}),$$

$$L = \frac{Z-1}{137.03} \sqrt{\frac{2m}{T_{\text{out}}(1 + T_{\text{out}}/2m)}} [\arccos x - x\sqrt{1-x^2}], \quad x = \sqrt{T_{\text{out}}/B} \quad (\text{Gamow factor}),$$

$$\text{free: } E_{\text{free}} = T_{\text{out}} + m_t, \quad \vec{p}_{\text{free}} = \hat{p}' \sqrt{E_{\text{free}}^2 - m_t^2} \quad (\text{no refraction}).$$

A reflection only delays the exit (the particle bounces and tries again), so the toy takes every $T_{\text{out}} > 0$ proton out: 98 % of the phase-space protons; $\langle P_T \rangle = 0.997$, $P_T < 0.99$ for 3 % of them.

7 Closed forms

Energy. Inserting the balance condition into $\omega - T_{p'}$:

$$\begin{aligned} E_{\text{miss}} &= V_0 - T_{\text{ball}} - V(T_{p'}) \\ &= V_0 - T_{\text{ball}} \in [S, V_0] = [6.83, 45.0] \quad \text{for fast protons } (V = 0, T_{p'} \gtrsim 195 \text{ MeV}). \end{aligned}$$

Independent of the local-energy setting: v_{loc} moves the scattering nucleon and the lepton, never E_{miss} . Slow protons keep $V(T_{p'}) > 0$ and reach E_{miss} down to $-T_F = -38$ MeV. Verified: $\leq 2.1 \times 10^{-12}$ MeV in all four runs.

Momentum. In the CM the two momenta cancel for every α , so the lab total after the boost is $\gamma\vec{\beta} E_{\text{tot}}^*(\alpha)$; before scaling it was $\vec{k} + \vec{p}_{\text{red}} = \gamma\vec{\beta}\sqrt{s}$. The energy condition fixes $\gamma[E_{\text{tot}}^*(\alpha) - \sqrt{s}] = V(T_{p'}) - V_0 + v_{\text{loc}}$, hence

$$\begin{aligned} \vec{p}_{\text{miss}} &= \vec{p}_{\text{red}} + \vec{\beta} [V(T_{p'}) - V_0 + v_{\text{loc}}(r)] & (\text{never: } v_{\text{loc}} = 0, \vec{p}_{\text{red}} = \vec{p}_{\text{ball}}), \\ &= \vec{p}_{\text{red}} - \vec{\beta} (V_0 - v_{\text{loc}}) & \text{fast protons.} \end{aligned}$$

The rescaling therefore shifts the final-state momentum against the beam by $\beta_z(V_0 - v_{\text{loc}})$: mean 22 MeV/c (on, $\langle\beta_z\rangle = 0.72$, $\langle v_{\text{loc}}\rangle = 13.9$) and 32 MeV/c (never), the difference between the blue and green $|\vec{p}_{\text{miss}}|$ curves; the remnant absorbs it (review note item 3). Verified: $\leq 2.8 \times 10^{-12}$ MeV/c for fast protons in all four runs. For slow protons with local energy on, the fixed point $E_{p'} \leftarrow E_{p'} + v_{\text{loc}}$ re-adjusts $|\vec{p}'|$ after the boost and the closed form no longer applies (deviations up to 236 MeV/c, mean 18, for the 9 % of phase-space events with $V(T_{p'}) > 0$).

Free proton. With ω unchanged by the exit,

$$\begin{aligned} E_{\text{miss}}^{\text{free}} &= \omega - T_{\text{out}} = V_0 - T_{\text{ball}} - \Delta Q = T_F + S_p^{\text{real}} - T_{\text{ball}} \in [S_p^{\text{real}}, T_F + S_p^{\text{real}}] = [15.96, 54.13] \text{ MeV}, \\ \vec{p}_{\text{miss}}^{\text{free}} &= \vec{p}_{\text{free}} - \vec{q} = \vec{p}_{\text{miss}} + (|\vec{p}_{\text{free}}| - |\vec{p}'|) \hat{p}'. \end{aligned}$$

$V(T_{p'})$ cancels: whatever potential the proton still carries inside is paid at the wall, so the negative tail of the record's E_{miss} disappears and the floor moves from INCL's $S = 6.83$ to the real $S_p = 15.96$ MeV, for every setting and every exiting proton (verified: $\leq 2.0 \times 10^{-12}$ MeV in all four runs; mean 31.1 MeV with the fuzzy ground state, 26.6 with **--resample**). The momentum loses $|\Delta Q|/\beta_{p'}$, i.e. 9.6–17 MeV/c along $\hat{p}'$ for fast protons, and $\langle|\vec{p}_{\text{miss}}|\rangle$ moves by 1–2 MeV/c.

8 Summary of the two settings

scattering nucleon P		$E_{\text{miss}}^{(0)}, \vec{p}_{\text{miss}}^{(0)}$	record E_{miss}	record $\vec{p}_{\text{miss}}$ (fast p')
on	$(E_{\text{ball}} - v_{\text{loc}}, \vec{p}_{\text{red}})$	$-T_{\text{red}}, \vec{p}_{\text{red}}$	$V_0 - T_{\text{ball}} - V(T_{p'})$	$\vec{p}_{\text{red}} - \vec{\beta}(V_0 - v_{\text{loc}})$
never	$(E_{\text{ball}}, \vec{p}_{\text{ball}})$	$-T_{\text{ball}}, \vec{p}_{\text{ball}}$	$V_0 - T_{\text{ball}} - V(T_{p'})$	$\vec{p}_{\text{ball}} - \vec{\beta}V_0$

free-proton E_{miss}		free-proton $\vec{p}_{\text{miss}}$
on and never	$T_F + S_p^{\text{real}} - T_{\text{ball}}$	$\vec{p}_{\text{miss}} + (\vec{p}_{\text{free}} - \vec{p}') \hat{p}'$

Numbers (200k events, 2.445 GeV, seed 1, **--resample**): $\langle |\vec{p}_{\text{red}}| \rangle = 147.8$ MeV/c with $\text{corr}(p_{\text{red}}, r) = -0.66$ (on) vs $\langle |\vec{p}_{\text{ball}}| \rangle = 225.5, +0.47$ (never); E_{miss} of fast protons: mean 17.46 in [6.83, 44.80] in both; $\langle |\vec{p}_{\text{miss}}| \rangle = 148.4$ (on) / 224.2 (never); free-proton E_{miss} mean 26.6 in [15.96, 53.9] in both, 98 % of the protons exit.